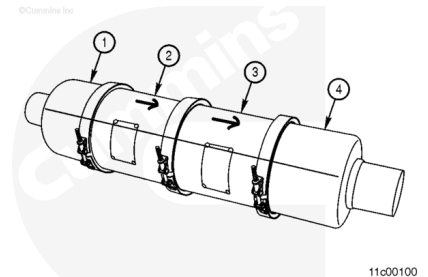


Trainings ▾

General Information

⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system as damage to the catalyst element can result.



Due to the number of exhaust aftertreatment applications, this procedure is generic. Illustrations within this procedure will **not** represent all applications.

The exhaust aftertreatment is composed of four sections. These sections are:

1. Inlet
2. Aftertreatment diesel oxidation catalyst (DOC)
3. Aftertreatment diesel particulate filter (DPF)
4. Outlet.

Note : In some applications the catalyst can be integrated into the inlet of the exhaust aftertreatment.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) batter cable last.



⚠ CAUTION ⚠

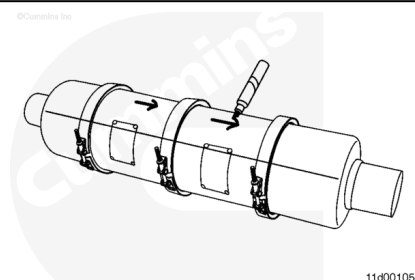
The catalyst can remain hot for a long time after the engine has stopped.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Disconnect the aftertreatment exhaust gas temperature sensor electrical connector(s) from the wiring harness, if necessary. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/101/101-019-449.html>) This procedure can be located in the Troubleshooting and Repair Manual, ISX and ISM Engines, Bulletin 4021560.
- Disconnect the aftertreatment diesel particulate filter differential pressure sensor tubes, if necessary. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html>)
- Disconnect the aftertreatment diesel particulate filter differential pressure sensor mounting bracket, and strap if applicable. Refer to Procedure 011-046 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-046-tr.html>)

Remove

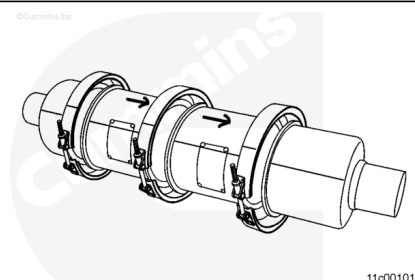
Draw an orientation reference line across each of the V-band clamps, aftertreatment canister sections, and connection points to the tailpipe. This will aid in returning the sections and V-band clamps to their original orientation during installation.

Mark the direction of exhaust flow on both the catalyst section and the aftertreatment diesel particulate filter section to aid in assembly.



⚠ WARNING ⚠

The component weighs 23 kg [50 lbs] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



⚠ CAUTION ⚠

The catalyst elements contained in the aftertreatment system are made of brittle material. Do not drop or strike the side of the aftertreatment system as damage to the catalyst element can result.

⚠ CAUTION ⚠

Do not use air tools to remove or install the nut on the V-band clamp. Use of these tools can seriously damage the threads or the bolt and cause the clamp to not be reusable.

Note : If necessary, remove additional mounting hardware to remove the catalyst from the vehicle.

Remove the V-band clamps from the inlet and outlet of the catalyst.

Separate the component sections by approximately one-half inch to allow removal over the gasket retainer rings.

Remove and discard the gaskets.

Clean and Inspect for Reuse

⚠ CAUTION ⚠

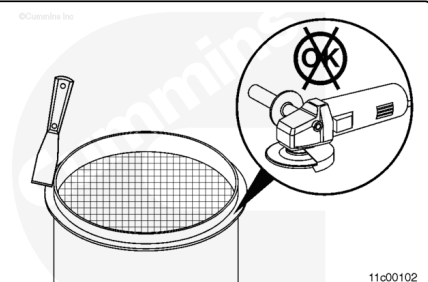
Do not use a grinder or abrasive air tool to remove residual gasket material, as this can damage the flange and cause the connection to leak.

⚠ CAUTION ⚠

Do not use an open flame to burn off soot accumulation from the face of the aftertreatment diesel particulate filter.

⚠ CAUTION ⚠

Do not scrape off soot accumulation from the face of the aftertreatment diesel particulate filter.



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Remove any residual gasket material from the flanges on the aftertreatment diesel particulate filter with a putty knife.

Avoid dropping fragments of gasket material into the aftertreatment diesel particulate filter.

Note : If the aftertreatment diesel particulate filter is being inspected due to progressive damage that introduced engine oil or excessive fuel into the exhaust, inspect the tailpipe from the turbocharger outlet to the aftertreatment DOC.

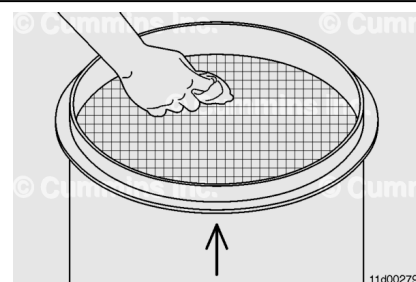
If a trail of liquid engine oil or fuel can be seen from the turbocharger outlet to the aftertreatment DOC, the aftertreatment DOC **must** be replaced.

If a trail of engine oil or fuel can be seen exiting the turbocharger outlet, the tailpipe between the turbocharger outlet and aftertreatment DOC **must** be cleaned. Refer to Procedure 011-048 in Section 11.

(/qs3/pubsys2/xml/en/procedures/101/101-011-048-tr.html)

⚠ WARNING ⚠

The material captured in a diesel particulate filter may contain elevated concentrations of metals, primarily zinc and molybdenum, and possibly polynuclear aromatic hydrocarbons that may be regulated. These materials must be characterized, handled, and disposed of according to applicable local regulations. In addition, due to the presence of the above-listed chemicals and other potentially toxic components such as oxides of calcium, zinc, phosphorous, silicone, sulfur, and iron, exhaust filter maintenance must be completed only by appropriately trained personnel.



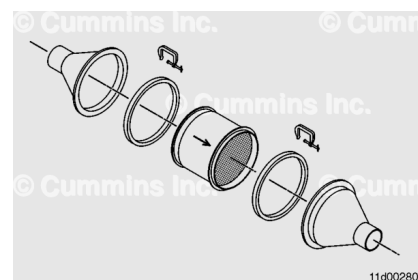
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the outlet of the aftertreatment DOC for loose debris and soot. Use a rag to wipe off any soot. Use a high-efficiency particulate air (HEPA) vacuum to remove any remaining loose debris.

Note : Use a vacuum bag, such as a drywall dust bag, to capture the soot that is removed.

Attach the Cummins® aftertreatment diesel particulate filter cleaning machine cone, Part Number 4918850, to the inlet and outlet sides of the aftertreatment DOC with the adapter rings, Part Number 4918847 and 4918848. Evenly space 4 to 5 spring clamps or C-clamps around the circumference of the aftertreatment DOC to seal the cone and adapter ring to the aftertreatment DOC.

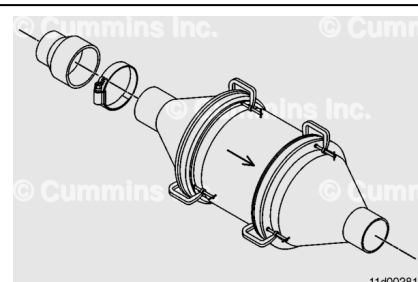


Service tool kit, Part Number 4918877 contains the cones and rings needed for this procedure.

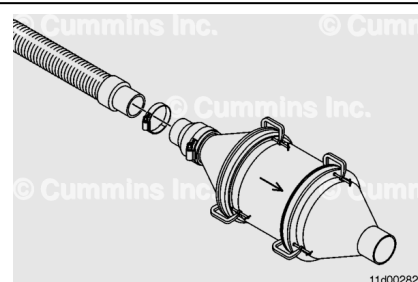
If the aftertreatment DOC is integrated into the aftertreatment inlet, use one cone only, and attach it to the outlet side of the unit.

Use tape to seal any open ports.

Obtain a 76 mm to 51 mm [3 in to 2 in] rubber pipe reducer from a local supplier. If the aftertreatment DOC is integrated into the aftertreatment inlet, obtain a 102 mm to 51 mm [4 in to 2 in] rubber pipe reducer.

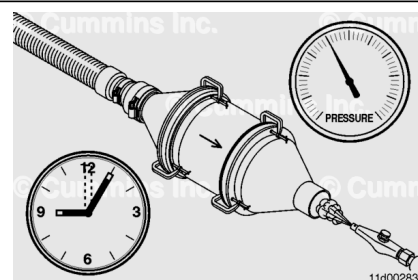


Attach the vacuum hose to the 51 mm [2 in] end of the pipe reducer. Attach the larger end of the reducer to the aftertreatment diesel particulate filter cleaning machine cone attached to the inlet side of the DOC. If the aftertreatment DOC is integrated into the inlet, install the larger end of the reducer onto the aftertreatment inlet.



⚠ CAUTION ⚠

**Use only a rubber-tipped air gun near the catalyst material.
Damage will result if a metal air gun strikes the catalyst material.**



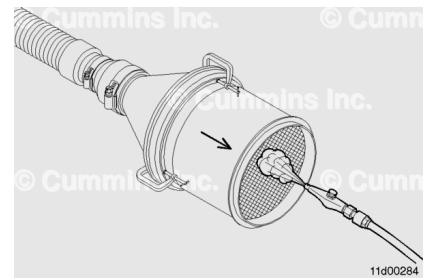
Adjust the air supply for a rubber-tipped air gun to 621 kPa [90 psi].

Switch on the vacuum. The vacuum and service tool kit is properly installed if the air flow through the aftertreatment DOC is opposite the exhaust flow. Exhaust flow is indicated by an arrow on the canister of the DOC.

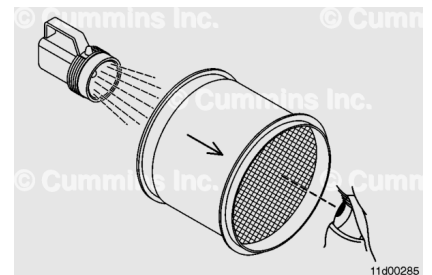
Blow compressed air across the outlet face of the aftertreatment DOC for approximately 5 minutes.

After 5 minutes, remove the aftertreatment diesel particulate filter cleaning machine cone and continue blowing compressed air into the outlet of the DOC for an additional 10 to 15 minutes. Maintain a 13 mm to 25 mm [1/2 in to 1 in] distance between the air gun and DOC face. Use a sweeping motion across the entire face of the DOC and attempt to blow air through every cell.

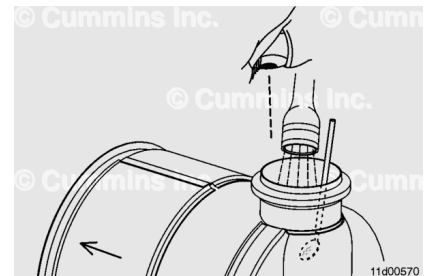
Allow the vacuum to operate for 1 minute after blowing out the aftertreatment DOC. After the vacuum is switched off, disconnect the cleaning machine cone and inspect the cells of the DOC.



Inspect the cells of the DOC by looking through each cell. Shine a light on the opposite side of the DOC to aid inspection. If the cell is clear of soot, it is possible to see straight through the DOC.

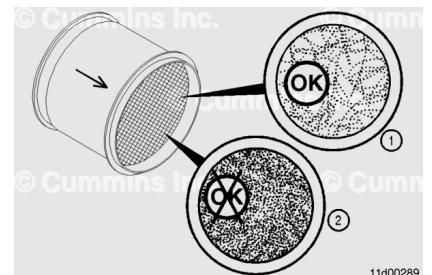


If the aftertreatment DOC has a side inlet configuration, insert a shop light and an inspection mirror into the inlet of the DOC to aid inspection.



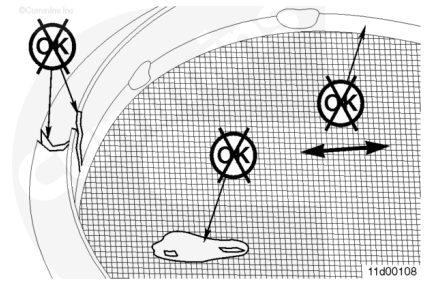
Upon inspection, if less than 50 percent of the cells are blocked (1) the aftertreatment DOC can be reused. If more than 50 percent of the cells remain blocked (2), repeat the cleaning procedure.

It should not take more than two attempts to clean the face of the DOC.



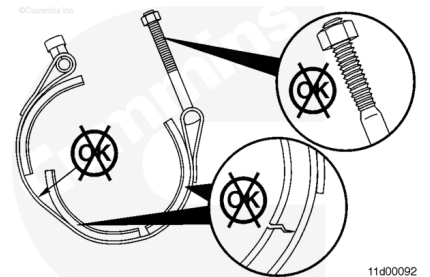
If the catalyst section has shifted, moved, or is loose inside the canister, replace the catalyst section.

Inspect the inlet and outlet faces of the aftertreatment DOC for signs of damage. Refer to Additional Aftertreatment DOC and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600 ([/qs3/pubsys2/xml/en/bulletin/4021600.html](http://qs3/pubsys2/xml/en/bulletin/4021600.html)).



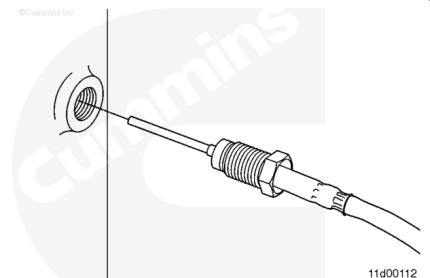
Inspect the V-band clamps and catalyst mounting straps for signs of over-extension. The V-band **must not** be bent or damaged.

Replace the V-band clamps or catalyst mounting straps if damage is found.



Install

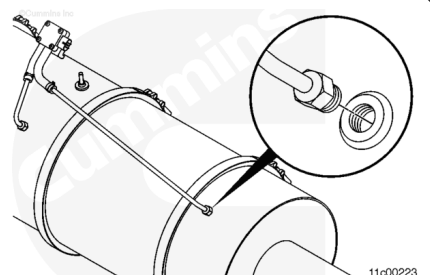
Install the exhaust gas temperature sensor into the catalyst section, if necessary. Refer to Procedure 019-449 in Section 19. ([/qs3/pubsys2/xml/en/procedures/101/101-019-449.html](http://qs3/pubsys2/xml/en/procedures/101/101-019-449.html)) This procedure can be located in the Troubleshooting and Repair Manual, ISX and ISM Engines, Bulletin 4021560.



Note : If the engine had turbocharger damage or any other damage that introduced foreign objects into the exhaust system, the aftertreatment exhaust gas temperature 1 sensor **must** be inspected for damage. If the sensor is broken or otherwise damaged, it **must** be replaced.

⚠ WARNING ⚠

The component weighs 23 kg [50 lbs] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.



Service locations in California. See Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787 ([/qs3/pubsys2/xml/en/bulletin/6642787.html](http://qs3/pubsys2/xml/en/bulletin/6642787.html)).

Install new gaskets on the inlet and outlet of the exhaust gas catalyst.

Install the catalyst.

Tighten the V-band clamps

Torque Value: 20 n•m [177 in-lb]

Install the aftertreatment diesel particulate filter differential pressure sensor tubes into the catalyst section.

Torque Value: 17 n•m [150 in-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) batter cable last.



- Connect the aftertreatment gas temperature sensor electrical connector(s) to the wiring harness, if necessary. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/101/101-019-449.html>)
- Connect the aftertreatment diesel particulate filter differential pressure sensor tubes, if necessary. Refer to Procedure 011-047 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html>)
- Connect the aftertreatment diesel particulate filter differential pressure sensor mounting bracket and strap if necessary. Refer to Procedure 011-046 in Section 11. (</qs3/pubsys2/xml/en/procedures/101/101-011-046-tr.html>)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine. Check for fault codes and exhaust leaks.

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>)

Last Modified: 26-Aug-2025
